

$$(f3)_D = (110101)_3 = (65)_D$$

$$(127)_D = (0111111)_B = (177)_O$$

$$(635)_{10} = (\underline{10011} \ \underline{1011})_2 \quad B = (1 \ 173)_8$$

$$(7.943)_D = (111.111)_3 = (7.7)_0$$

$$(7943)_D = (1001111.011)_B = (117.344)_{10}$$

$$635 = 512 + 128 - 5$$

$$2^9 + 2^7 - 5$$

$$\begin{array}{r} \cancel{27.943} \\ 0.943 \\ \hline 2 \end{array}$$

$$\begin{array}{r} 0.943 \\ \underline{2} \\ 1.886 \\ \underline{0.886} \\ 1.12 \\ \underline{1.12} \\ 0.772 \\ \underline{0.772} \\ 1.544 \end{array}$$

$$2. (1010)_3 = (12)_6 = (10)_D$$

$$(11101)_3 = (75)_9 = \cancel{(79)}(61)_{10}$$

$$(101100)_2 = (134)_8 = (92)_{10}$$

$$(0.\underline{100110})_3 = (0.46)_6 = \left(\frac{23}{50} \right)_{10}$$

$$(\underline{0111})_B = \cancel{15}_D = (47)_D$$

$$(01101)_B = (15)_O = (13)_D$$

$$V = \frac{1}{2} + 2^{-4} + 2^{-5}$$

$$3. (1997)_D = \cancel{(01111100101)}_2 = 0001\ 1001\ 1001\ 0111$$

$$(6J312)_D = \cancel{1000000} \cancel{01000001} = \cancel{2}^{10} 011.0011 \text{ and } 0010$$

$$(3.1416)_D = \cancel{(11.0010)}_B = 0011.0001\ 0100\ 0001\ 0110$$

$$(0.9475)_D = (\underline{2.1111})_3 = 0.1001 \mid 0.1001 \mid 0.11 \mid 0.101$$

4.

A	B	C	X
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

~~$X = A^T A + I$~~

$$X = ABC + AC\bar{B} + \bar{A}BC$$

7. (i) $A + \bar{A}B = (A + \bar{A})(A + B) = 1 \cdot (A + B) = A + B$

$$(2) \quad ABC + A\bar{B}C + AB\bar{C} = A(BC + \bar{B}C + B\bar{C}) = A(B(C + \bar{C}) + \bar{B}C) = A(B + \bar{B}C) = A(B + C)$$

$$(3) A + \cancel{A}\bar{B}\bar{C} + \bar{A}CD + (\bar{C} + \bar{D})\bar{E} = A + CD + \bar{A}\bar{B}\bar{C} + (\bar{C} + \bar{D})\bar{E} = A + CD + (\bar{C}\bar{D})\bar{E} = A + CD + \bar{E}$$



$$(14) \bar{A}\bar{B} + A\bar{B}C + \bar{A}BC + AB(\quad)$$

$$A\bar{B} + ABC$$

$$(A\bar{B} + AB + AB)$$

$$(A\bar{B})$$

$$\bar{A}\bar{B} + \bar{A}\bar{C} + \bar{B}\bar{C}$$

$$8. (1) F: A + B(C + D)\bar{B}(A + \bar{C})$$

$$A + BC + \bar{B}\bar{C} = A + B\oplus C$$

$$(2) F = AC(\bar{B}D + \bar{B}) + A\bar{B} + A\bar{D}(H\bar{B}\bar{C})$$

$$= AC\bar{B} + A\bar{B} + A\bar{D}$$

$$= A\bar{B} + A\bar{D}$$

$$(3) F = AB(C\bar{D} + CD + D) + B(C\bar{D} + \bar{C})$$

$$= ABC(C\bar{D} + CD + D) + B(\bar{C} + \bar{D})$$

$$= ABCD + ABC\bar{C} + ABD + B\bar{C}\bar{D}$$

$$= B(ACD + \bar{C}) + A\bar{B}C + A\bar{B}D$$

$$= \underline{AB + B\bar{C} + \bar{B}D + A\bar{B}C + A\bar{B}D} = AB + B\bar{C} + B\bar{D}$$

$$(4) F = \bar{A}\bar{C} \cdot \bar{A}\bar{B}C + \bar{B}\bar{C} + A\bar{B}\bar{C}$$

$$= (\bar{A}\bar{C} + \bar{A}\bar{B}C)(\bar{B}\bar{C}) + (\bar{A} + \bar{B} + \bar{C})$$

$$= (\bar{A}\bar{B}C + \bar{B}\bar{C})(\bar{A} + \bar{B} + \bar{C})$$

$$= \bar{B}\bar{C} + \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}\bar{C} = \bar{B}\bar{C}$$

$$9. (1) F = A + (\bar{B} + \bar{C})$$

$$= A + \bar{B}\bar{C}$$

$$= A(B + \bar{B})(C + \bar{C}) + \bar{B}\bar{C}(A + \bar{A})$$

$$= (AB + A\bar{B})(C + \bar{C}) + A\bar{B}C + \bar{A}\bar{B}\bar{C}$$

$$= \underbrace{ABC}_{m_7} + \underbrace{A\bar{B}C}_{m_6} + \underbrace{A\bar{B}\bar{C}}_{m_5} + \underbrace{\bar{A}\bar{B}\bar{C}}_{m_4} + \underbrace{\bar{A}\bar{B}C}_{m_2} + \underbrace{\bar{A}\bar{B}\bar{C}}_{m_1} = \sum m(1, 2, 4, 5, 6, 7)$$

$$F = (\bar{A}\bar{B})(\bar{A} + \bar{B} + \bar{D})(B + \bar{C}D)$$

$$= (\bar{A}\bar{B} + \bar{A}\bar{D} + \bar{A}\bar{B} + \bar{B}\bar{D})(B + \bar{C}D)$$

$$= \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}\bar{D} + \bar{A}\bar{B} + \bar{A}\bar{B}\bar{C}D + \bar{B}\bar{D}$$

$$= \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}\bar{C}\bar{D}$$

$$= \sum m(5, 6, 7, 9, 12, 14)$$



$$\begin{aligned}
 10. (1) F &= (\bar{A} + \bar{C})(A + B\bar{C})(B + \bar{C}) + A\bar{B}\bar{C} \\
 &= (\bar{A}\bar{B} + \bar{A}\bar{C} + A\bar{C} + \bar{B}\bar{C} + \bar{C})(B + \bar{C}) + A\bar{B}\bar{C} \\
 &= \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}\bar{C} + \bar{A}\bar{C} + A\bar{B}\bar{C} + A\bar{C} + \bar{B}\bar{C} + \bar{C} + A\bar{B}\bar{C}
 \end{aligned}$$

A \ BC	00	01	11	10
0	1	0	0	1
1	1	0	0	1

$$F = \bar{C}$$

(2)

AB \ CD	00	01	11	10
00	0	0	0	0
01	0	0	0	0
11	1	0	0	1
10	1	1	1	1

$$F = \bar{A}\bar{B} + A\bar{D}$$

(3)

AB \ CD	00	01	11	10
00	1	1	0	1
01	0	1	1	1
11	0	1	0	1
10	1	1	0	0

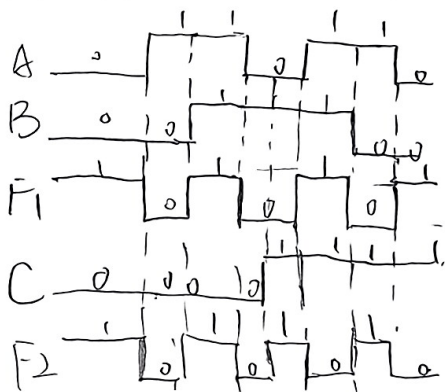
$$F = \bar{B}\bar{C} + \bar{C}\bar{D} + \bar{A}B\bar{C} + \bar{A}\bar{C}\bar{D} + B\bar{C}\bar{D}$$

(4)

AB \ CD	00	01	11	10
00	1	0	0	0
01	0	0	0	0
11	0	1	1	1
10	0	1	1	1

$$F = \bar{A}\bar{B} + AD + AC$$

$$14. F_1 = \bar{A}B + A\bar{B} = A \oplus B \quad F_2 = F_1 \oplus C$$



$$1. F = AB + \bar{B} = \overline{AB \cdot B} = \overline{A \cdot \bar{B}} = \bar{A}B$$

$$F = \overline{AB \cdot B \cdot \bar{B} \cdot C \cdot \bar{B} \cdot C} = \bar{A}B + B\bar{A}C + C \cdot \bar{A}\bar{B}C = \bar{A}B + B(\bar{A} + \bar{B} + \bar{C}) + C(\bar{A} + \bar{B} + \bar{C})$$

$$= \bar{A}B + \bar{B}C + \bar{A}C + \bar{B}C$$

$$= m_1 + m_2 + m_3 + m_6$$

$$5. F_1 = x_0 \bar{x}_1 \bar{x}_0$$

$$F_2 = x_1 \bar{x}_1 \bar{x}_0$$

$$F_3 = x_2 \bar{x}_1 \bar{x}_0$$

$$F_4 = x_3 \bar{x}_1 \bar{x}_0$$

$$F = F_1 + F_2 + F_3 + F_4 = x_0 \bar{x}_1 \bar{x}_0 + x_1 \bar{x}_1 \bar{x}_0 + x_2 \bar{x}_1 \bar{x}_0 + x_3 \bar{x}_1 \bar{x}_0$$

A \ BC	00	01	10	11
0		1	1	1
1		1		

$$F = \bar{B}C + \bar{A}B + \bar{B}C$$

选择输入		输出
A ₁	A ₀	F
0	0	$\bar{x}_0$
0	1	x_1
1	0	x_2
1	1	x_3

四选一数据选择器

9. 设三位数分别为ABC, Y_G表示绿灯, Y_R表示红灯, Y_Y表示黄灯, 0表示故障, 1表示正常工作

	A	B	C	Y _G	Y _R	Y _Y
m ₀	0	0	0	0	0	1
m ₁	0	0	1	0	1	0
m ₂	0	1	0	0	1	0
m ₃	0	1	1	1	0	0
m ₄	1	0	0	0	1	0
m ₅	1	0	1	1	0	0
m ₆	1	1	0	1	0	0
m ₇	1	1	1	1	1	0

$$Y_G = \bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B}\bar{C} + A\bar{B}C = A(\bar{B}C + B\bar{C}) + \bar{A}C$$

$$= A(B \oplus C) + \bar{A}C$$

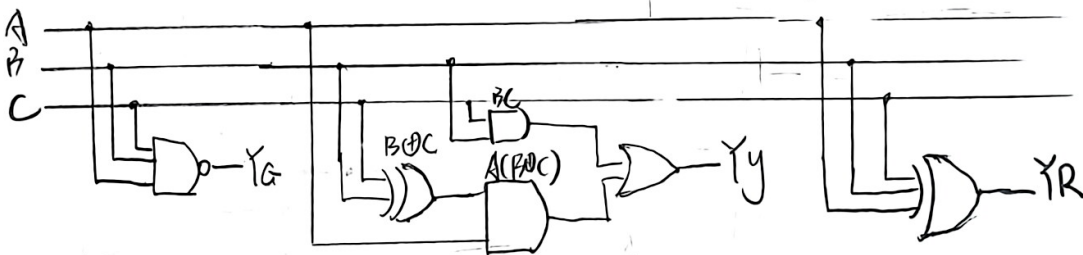
$$Y_R = \bar{A}BC + \bar{A}\bar{B}C + A\bar{B}\bar{C} + A\bar{B}C = A(\bar{B}C + B\bar{C}) + \bar{A}C$$

$$= A(B \oplus C) + \bar{A}C$$

$$= A(\bar{B}C) + \bar{A}(B \oplus C)$$

$$= A \oplus B \oplus C$$

$$Y_Y = \bar{A}\bar{B}\bar{C} = \overline{A + B + C}$$



AB 表示两位型 (00 表示 00) G 表示 13 灯 R 表示 14 灯

00 A
01 B
10 AB
11 0

1 亮
0 不亮

18

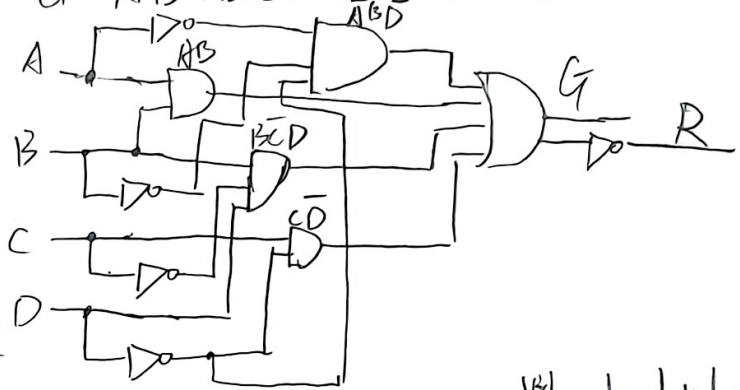
	A	B	C	D	G	R
A → A	0	0	0	0	1	0
A → B	0	0	0	1	0	1
A → AB	0	0	1	0	1	0
A → 0	0	0	1	1	0	1
B → A	0	1	0	0	0	1
B → B	0	1	0	1	1	0
B → AB	0	1	1	0	1	0
B → 0	0	1	1	1	0	1
C → A	1	0	0	0	0	1
C → B	1	0	0	1	0	1
C → AB	1	0	1	0	1	0
C → 0	1	0	1	1	0	1
D → A	1	1	0	0	1	0
D → B	1	1	0	1	1	0
D → AB	1	1	1	0	1	0
D → 0	1	1	1	1	1	0

$$G = \sum m(0, 2, 3, 6, 10, 12, 13, 14, 15)$$

$$R = \bar{G}$$

CD \ AB	00	01	11	10
00	1			1
01		1		1
11	1	1	1	1
10				1

$$G = AB + B\bar{C}D + C\bar{D} + \bar{A}\bar{B}\bar{D}$$



BC \ A	00	01	11	10
0				
1		1	1	1

AC + AB

$$= m_5 + m_6 + m_7$$

$$(2) F = ABC + ABC + ABC$$

$$= AC + AB$$

$$= AC + AB = \overline{AC} \cdot \overline{AB}$$

$$= m_1 + m_2 + m_3$$

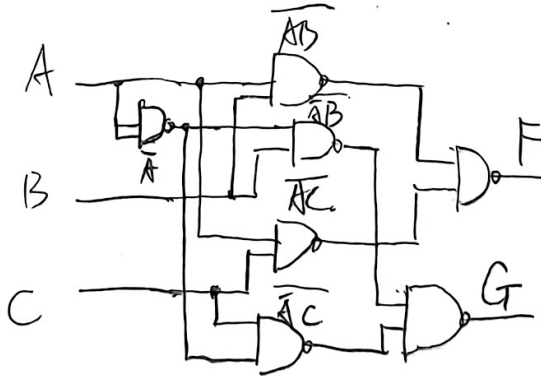
$$G = \bar{A}\bar{B}C + \bar{A}B\bar{C} + \bar{A}BC$$

$$= \bar{A}B + \bar{A}\bar{C}$$

$$= \bar{A}B + \bar{A}\bar{C} = \bar{A}B \cdot \bar{A}\bar{C}$$

$$\bar{A}B + \bar{A}\bar{C}$$

(3) A



第3章作业

1. 当 $\bar{a}=0$ 时, $\bar{Q}=1$, $Q=0$ 置 0

当 $\bar{b}\bar{c}=0$ 时, $Q=1$, $\bar{Q}=0$ 置 1

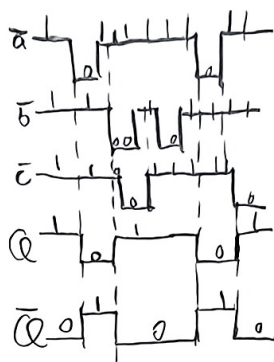
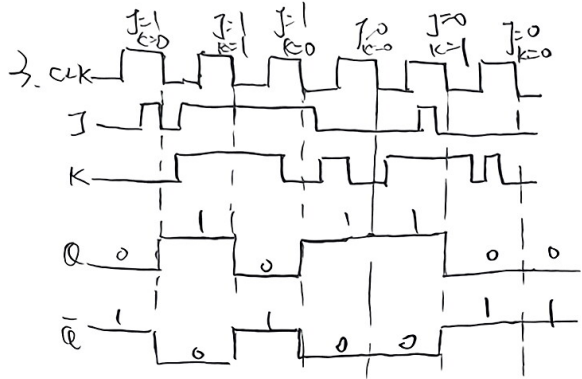
当 $\bar{a}=\bar{b}\bar{c}=1$ 时, $\bar{Q}=\bar{Q}$, $Q=Q$, 保持

$$\bar{a} = \bar{R}$$

$$\bar{b}\bar{c} = \bar{S}$$

$$Q^{n+1} = \bar{b}\bar{c} + \bar{a}Q^n$$

$$= (b+c) + \bar{a}Q^n$$

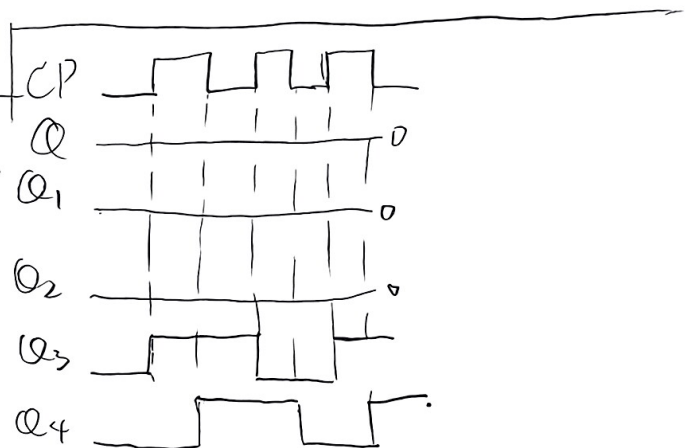


5. $Q_1^{n+1} = D = Q^n \text{ clk} \uparrow$

$Q_2^{n+1} = J\bar{Q}_n + \bar{K}Q_n = Q_n\bar{Q}_n + Q_n = Q_n \text{ clk} \downarrow$

$Q_3^{n+1} = D = \bar{Q}_n \text{ clk} \uparrow$

$Q_4^{n+1} = \bar{Q}_n\bar{Q}_n + 0 = \bar{Q}_n \text{ clk} \downarrow$



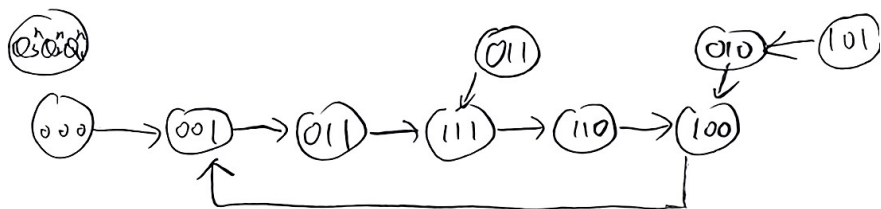
7. $\text{clk} \downarrow$

$J_1 = \bar{Q}_2^n \quad J_2 = Q_1^n \quad J_3 = Q_2^n$
 $K_1 = Q_3^n \quad K_2 = \bar{Q}_1^n \quad K_3 = \bar{Q}_2^n$

$Q_1^{n+1} = \bar{Q}_2^n\bar{Q}_1^n + Q_2^nQ_1^n$

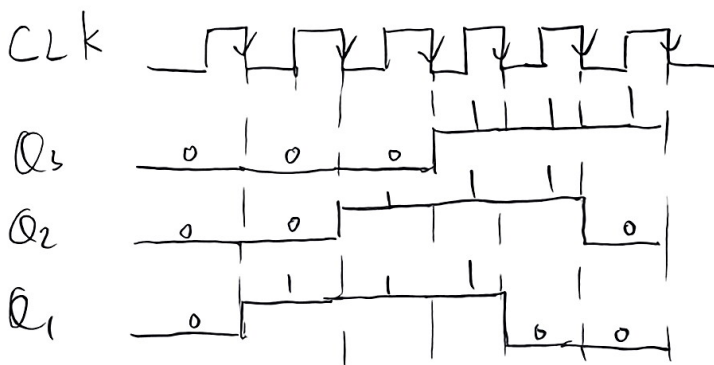
$Q_2^{n+1} = \bar{Q}_1^n\bar{Q}_2^n + Q_1^nQ_2^n = Q_1^n$

$Q_3^{n+1} = Q_2^n\bar{Q}_3^n + Q_3^nQ_2^n = Q_2^n$



五进制计数器 可自启动

PS	NS	
$Q_3^n Q_2^n Q_1^n$	$Q_3^{n+1} Q_2^{n+1} Q_1^{n+1}$	
0 0 0	0 0 1	✓
0 0 1	0 1 1	✓
0 1 0	1 0 0	✓
0 1 1	1 1 1	✓
1 0 0	0 0 1	✓
1 0 1	0 1 0	✓
1 1 0	1 0 0	✓
1 1 1	1 1 0	✓



9 由分析求用3个触发器

PS	NS	次方	
$Q_3^n Q_2^n Q_1^n$	$Q_3^{n+1} Q_2^{n+1} Q_1^{n+1}$	$D_3 D_2 D_1$	Σ
0 0 0	0 0 1	0 0 1	0 m_0
0 0 1	0 1 1	0 1 1	0 m_1
0 1 1	1 1 1	1 1 1	0 m_3
1 1 1	1 0 1	1 0 1	0 m_7
1 0 1	1 0 0	1 0 0	0 m_5
1 0 0	0 0 0	0 0 0	1 m_4
0 1 0	$\varnothing \varnothing \varnothing$	$\varnothing \varnothing \varnothing$	$\varnothing m_2$
1 1 0	$\varnothing \varnothing \varnothing$	$\varnothing \varnothing \varnothing$	$\varnothing m_6$

	Q_3^n	Q_2^n	Q_1^n	
D_3	0	0	1	1
	1	1	1	1
D_2	0	1	1	1
	1	1	1	1
D_1	0	1	1	1
	1	1	1	1
Σ	0	1	1	1
	1	1	1	1
Q_3^{n+1}	0	0	1	1
	1	1	1	1
Q_2^{n+1}	0	1	1	1
	1	1	1	1
Q_1^{n+1}	0	1	1	1
	1	1	1	1

$$D_3 = Q_2^n + Q_3^n Q_1^n \checkmark$$

$$D_2 = \overline{Q_3^n} Q_1^n \checkmark$$

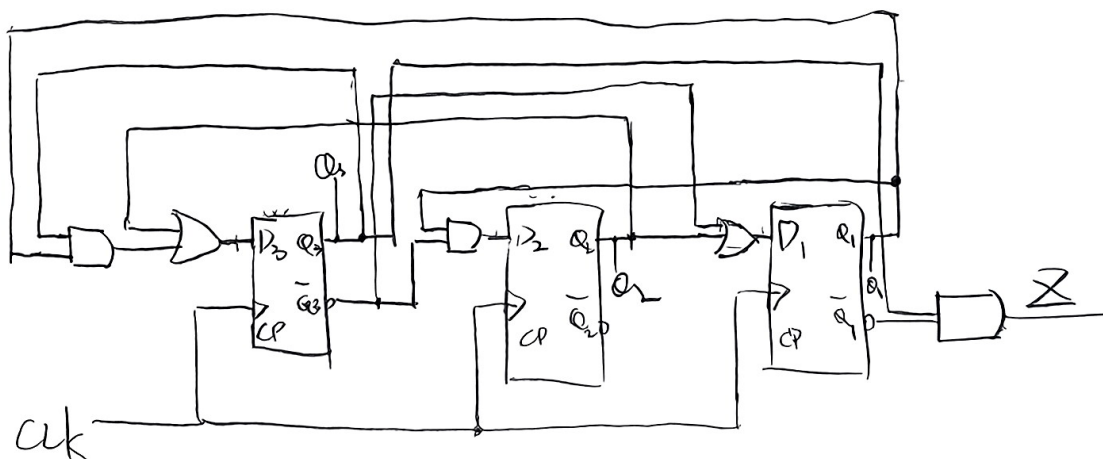
$$D_1 = \overline{Q_3^n} + Q_2^n \checkmark$$

$$\Sigma = Q_3^n \overline{Q_1^n}$$

$$Q_3^{n+1} = Q_2^n + Q_3^n Q_1^n$$

$$Q_2^{n+1} = \overline{Q_3^n} Q_1^n$$

$$Q_1^{n+1} = \overline{Q_3^n} + Q_2^n$$



13. clk ↓

$$J_1 = 1 \quad J_2 = Q_1^n \oplus X = K_2$$

kc = 1

$$Q_1^{n+1} = \overline{Q_1^n} \quad Q_2^{n+1} = (Q_1^n \oplus X) \overline{Q_2^n} + (\overline{Q_1^n \oplus X}) Q_2^n$$

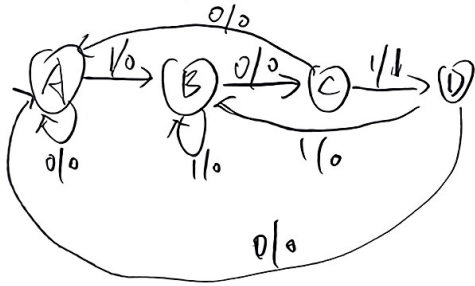
$$Z = \overline{Q_1^n Q_2^n}$$

$Q_2^n Q_1^n$	$Q_2^{n+1} Q_1^{n+1} / Z$	
	$X=0$	$X=1$
0 0	0 1 / 1	1 1 / 1
0 1	1 0 / 1	0 0 / 1
1 0	1 1 / 1	0 1 / 1
1 1	0 0 / 0	1 0 / 0



* 逻辑型
X=1 时，模4减法
X=0 时，模4加法

15.



X	X=0	X=1
A	A/0	B/0
B	C/0	B/0
C	A/0	D/1
D	A/0	B/0

17. 顺序、状态

A	C, 1	D, 1
B	B, 0	C, 1
C	C, 1	A, 0
D	D, 0	C, 0

A=00
B=01
C=10
D=11

$Q_2^n Q_1^n$	$X=0$	$X=1$
0 0	1 0 / 1	1 1 / 1
0 1	0 1 / 0	1 0 / 1
1 0	1 0 / 1	0 0 / 0
1 1	1 1 / 0	1 0 / 0

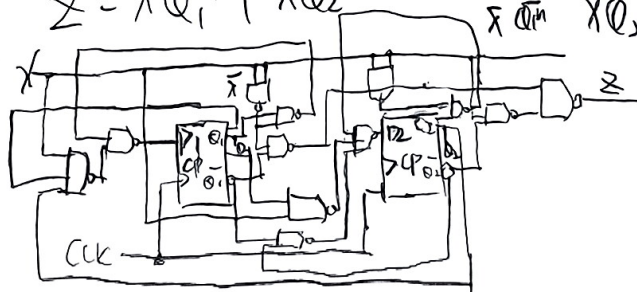
$Q_2^{n+1} Q_1^{n+1} / Z$		
$Q_2^n Q_1^n$	$X=0$	$X=1$
0 0	1 0 / 1	1 1 / 1
0 1	0 1 / 0	1 0 / 1
1 0	1 0 / 1	0 0 / 0
1 1	1 1 / 0	1 0 / 0

$X \backslash Q_2^n Q_1^n$	00	01	11	10
$D_1 = Q_2^{n+1}$	1	1	1	1
$D_2 = Q_1^{n+1}$	1	1	1	1
Z	1	1	1	1

$$D_1 = \overline{X} Q_1^n + X \overline{Q_2^n} \overline{Q_1^n} = \overline{X} \overline{Q_2^n} + X \overline{Q_2^n} \overline{Q_1^n}$$

$$D_2 = \overline{X} Q_2^n + X \overline{Q_2^n} + \overline{Q_2^n} \overline{Q_1^n} = \overline{X} \overline{Q_2^n} + X \overline{Q_2^n} + \overline{Q_2^n} \overline{Q_1^n}$$

$$Z = \overline{X} \overline{Q_1^n} + X \overline{Q_2^n} = \overline{X} \overline{Q_1^n} + X \overline{Q_2^n}$$



19 $Q_2^n Q_1^n$
 $S_0 = 00$
 $S_1 = 01$
 $S_2 = 10$
 $S_3 = 11$

Q_2^n	Q_1^n	Q_2^{n+1}	Q_1^{n+1}
0	0	0	0
0	1	0	1
1	0	1	1
1	1	1	0

14/6 奇数寄存器

Q_2^{n+1}

$Q_2^n Q_1^n$	00	01	11	10
0			0	1
1	1			1

Q_1^{n+1}

$Q_2^n Q_1^n$	00	01	11	10
0		1	1	1
1	1			

$$D_2^{n+1} = \bar{k} Q_1^n + Q_2^n \bar{Q}_1^n + k \bar{Q}_2^n Q_1^n$$

$$D_1^{n+1} = k \bar{Q}_2^n \bar{Q}_1^n + \bar{k} Q_1^n + \bar{k} Q_2^n$$

